

SIS-211 Introduction to DB Systems

Phase 2: Case Study ERD

Renewable Energy Grid & Maintenance Network

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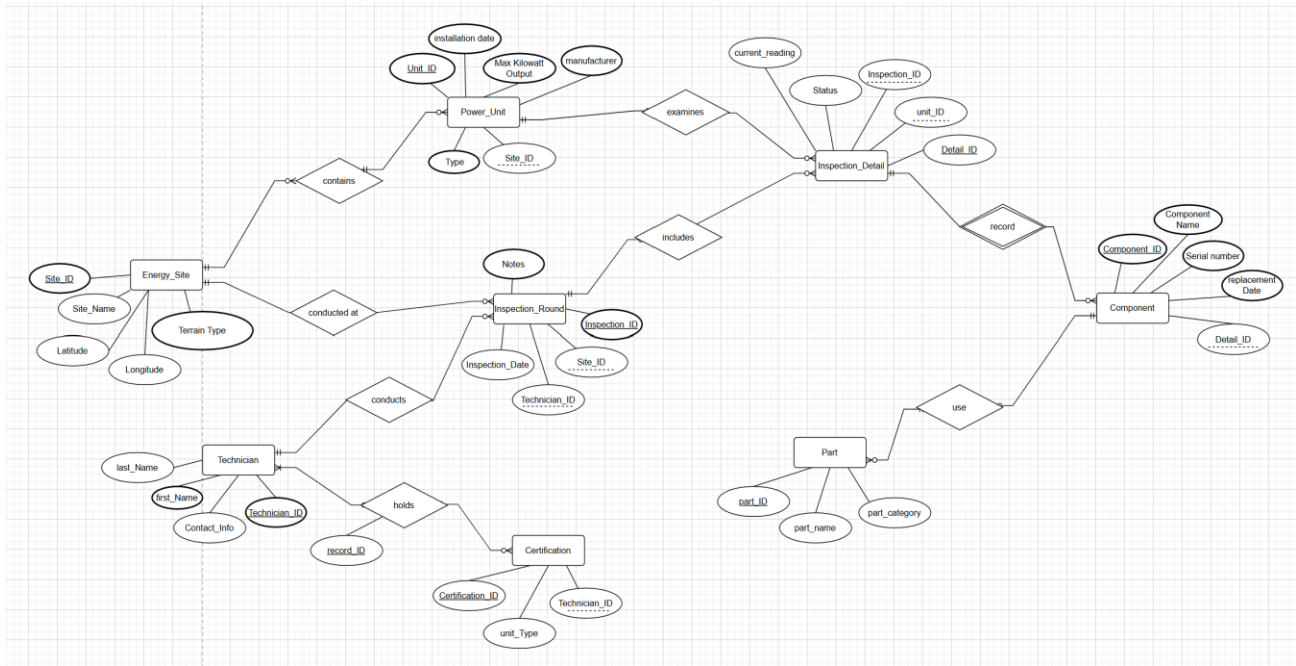
1. Functional Requirements

ID	Category	Description
FR-01	Dashboard — summary	The system shall display real-time counts of all entities: energy sites, power units, technicians, inspection rounds, components, and spare parts.
FR-02	Dashboard — recent	The system shall show the 10 most recent inspection rounds, joined with the corresponding site name and technician full name.
FR-03	Energy sites — CRUD	Users shall be able to create, read, update, and delete energy site records. Each record stores a site name, latitude, longitude, and terrain type.
FR-04	Energy sites — search	Users shall be able to search and filter energy sites by site name or terrain type in real time without a page reload.
FR-05	Power units — CRUD	Users shall be able to create, read, update, and delete power unit records linked to an energy site. Each record stores the unit type, manufacturer, maximum kW output, and installation date.
FR-06	Power units — filter	Users shall be able to filter power units by manufacturer or unit type.
FR-07	Technicians — CRUD	Users shall be able to create, read, update, and delete technician profiles storing first name, last name, and contact information.
FR-08	Technicians — certifications	Users shall be able to add and remove unit-type certifications for individual technicians directly from the technician list view.
FR-09	Technicians — search	Users shall be able to search technicians by name or contact information.
FR-10	Inspection rounds — CRUD	Users shall be able to create, read, update, and delete inspection rounds. Each round is linked to an energy site and a technician, and records an inspection date and optional notes.
FR-11	Inspection details — CRUD	Users shall be able to log, edit, and delete per-unit inspection detail records within a round, capturing operational status and current kW reading.
FR-12	Components — CRUD	Users shall be able to create, read, update, and delete component records linked to a power unit and an inspection detail. Each record stores component name, serial number, and replacement date.
FR-13	Spare parts — CRUD	Users shall be able to create, read, update, and delete spare part records linked to a component, with a category (Mechanical, Electrical, Hydraulic, Electronic, Structural, Safety, Sensor, Other).
FR-14	Analytics — below average	The system shall identify and rank manufacturers by the number of power unit readings that fall below the database-wide average kW reading.
FR-15	Analytics — no inspections	The system shall list all energy sites that had no inspection round recorded in the previous calendar month.
FR-16	Analytics — top technician	The system shall rank technicians by the number of inspection rounds completed in the previous calendar month.
FR-17	Analytics — components/site	The system shall list all components installed at each energy site during the previous calendar month.
FR-18	Analytics — technician profile	The system shall show each technician alongside the total number of distinct power units they have inspected across all rounds.
FR-19	UI — confirm on delete	All delete operations shall require confirmation via a modal confirmation dialog before the record is removed.

2. Non-Functional Requirements

- **NFR-1: Data Integrity & Consistency:** The system must implement strict validation rules to ensure accuracy and prevent redundancy across the relational database.
- **NFR-2: High-Performance Querying:** The database architecture must be optimized to support efficient complex joins and rapid execution of analytical queries for reporting.
- **NFR-3: Scalability:** The system design shall be capable of horizontally scaling to accommodate an increasing volume of energy sites, assets, and high-frequency sensor readings.
- **NFR-4: Security & Access Control:** The system must ensure secure data handling, restricting access to sensitive technician profiles and maintenance logs through appropriate authorization levels.
- **NFR5: Auditability & Historical Archiving:** The system shall maintain a robust historical archive of all inspections and component changes to support long-term trend analysis and compliance audits.

3. ERD(Conceptual Design)



4. ERD(Physical Design)

